RUDRA PATEL

rudrapatel20@gmail.com LinkedIn - drrudrapatel (214) 566-3046

PROFESSIONAL SUMMARY:

Data Solutions Developer with a Master's in Health Informatics and 4 years of experience building secure RESTful APIs, ETL pipelines, and automated data integrations. Expert in bridging the gap between clinical research requirements and technical architecture using Python, Node.js, and Informatica. Specialized in REDCap API integration and clinical data governance.

TECHNICAL SKILLS:

- **Languages:** Python (ETL automation, data validation), R, Java, C#, SQL (PL/SQL), UNIX Shell Scripting.
- **Data Integration:** Informatica Intelligent Cloud Services (IICS), REDCap API, RESTful Web Services.
- **Web/Backend:** Node.js, Express.js, PHP, XML, jQuery, OpenShift, Joomla, WordPress.
- **Databases:** PostgreSQL, Oracle, MySQL, NoSQL, Neo4j, SQL Server.
- **Analytics:** Tableau, PowerBI, SAS, Geospatial Data Analysis.
- **Infrastructure:** AWS, Azure, Windows Server Cluster, Active Directory/LDAP, Qualys.

PROFESSIONAL EXPERIENCE:

Data Solutions Developer
University of North Carolina at Chapel Hill, Chapel Hill, NC Feb 2022 - Present

- Design, develop, and maintain secure RESTful APIs using Node.js/Express, integrating data from PostgreSQL, Oracle, and MySQL databases to support internal research and administrative applications.
- Build and optimize ETL pipelines using Informatica (IICS) and Python-based ETL scripts to extract, transform, and load data from heterogeneous clinical, financial, and IRB-related data sources.
- Develop and maintain REDCap integrations, including REDCap API-based automated imports, data validation, recoding, and post-production support for multiple clinical research projects.
- Integrate LDAP and Active Directory for user identity resolution and secure system access across internal applications.
- Support and analyze IRBIS data and Clinical Data Warehouse datasets to enable reporting, compliance, and research analytics.
- Perform advanced SQL and PL/SQL tuning and database optimization to reduce load times and improve query performance.
- Design and deliver Tableau and PowerBI dashboards, translating complex healthcare datasets into actionable insights.
- Support applications and services deployed in Red Hat OpenShift, Microsoft Azure and Windows Server/cluster environments, collaborating with infrastructure and security teams.
- Utilize Qualys vulnerability data to support security reporting and remediation tracking.
- Contribute to and champion enterprise-wide data governance initiatives, including metadata management, lineage, data quality monitoring, and standardized data access practices.
- Manage Joomla CMS deployments on Red Hat OpenShift, supporting containerized hosting, configuration updates, security patching, and operational stability.
- Work with geospatial datasets to support location-based analysis and visualization for research and institutional projects.
- Implement creative & efficient solutions to clean, join, and analyze the data while adhering to firm-wide data use guidelines.

Scientific Informatics Intern
Agios Pharmaceuticals, Cambridge, MA Jun - Aug 2021

- Built data pipelines to store experimental and clinical trial data in data warehouse using Rest API in C# and Azure Web Services.
- Designed schemas and developed connectivity to heterogeneous data sources using SQL Server Management Studio (SSMS).
- Built an automated notification system and developed PowerBI dashboards to track KPIs and deliver insights.

PUBLICATIONS:

Assessing Hospital Resource Utilization with Application to Imaging for Patients Diagnosed with Prostate Cancer
Healthcare | **doi.org/10.3390/healthcare10020248**

- Analyzed over- and under-utilization of inpatient imaging resources using R.
- Applied Multiple Logistic Regression, Poisson Rate Regression, Decision Curve & Tree Analysis, and Network Analysis.
- Identified 17.48% imaging misutilization, with statistically significant variation by age and race.

EDUCATION:

Master of Science in Health Informatics & Analytics GPA: 4.0/4.0
The University of North Carolina at Charlotte, Charlotte, NC
Bachelor of Dental Surgery (B.D.S.) GPA: 3.8/4.0
Geetanjali Dental & Research Institute, India